The One-Step Subspace in the Quantum Wielandt Inequality

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1 The assertion

The quantum Wielandt inequality of Sanz–Pérez-García–Wolf–Cirac is stated in [SPGWC10, Theorem 1].[1] Let $\mathcal{E}_A$ be a quantum channel on $M_D(\mathbb{C})$ with Kraus operators $A_1, \dots, A_a$. For each n, set

$$S_n(A) = \operatorname{span}\{A_{i_1} \cdots A_{i_n} : 1 \le i_1, \dots, i_n \le a\}.$$

Thus $S_n(A)$ is the subspace generated by words of length exactly n. Let

$$i(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\},$$

when this minimum exists.

The paper writes d for the number of linearly independent Kraus operators. Equivalently, after replacing the displayed Kraus list by a minimal spanning list, one has

$$d = \dim S_1(A).$$

With this convention, Theorem 1 asserts that a primitive channel satisfies

$$i(A) \le (D^2 - d + 1)D^2.$$

The same theorem also gives two better estimates under additional hypotheses on the one-step subspace. If $S_1(A)$ contains an invertible matrix, then

$$i(A) \le D^2 - d + 1.$$

If $S_1(A)$ contains a non-invertible matrix with a nonzero eigenvalue, then

$$i(A) \le D^2.$$

These are hypotheses about arbitrary matrices in $S_1(A)$. The paper does not require the special matrix to be one of the Kraus operators in a chosen Kraus representation.

[1] For traceability, this note corresponds to issue #1049. The local source passage is `Papers/0909.5347/main.tex:360--365`, `Papers/0909.5347/main.tex:572--590`, and `Papers/0909.5347/main.tex:645--684`.

2 What has been formalized

The formal statements use the same subspaces $S_n(A)$ and define the Kraus rank by

$$\mathrm{kr}(A) = \dim S_1(A).$$

This is the paper's number d written invariantly. Replacing d by $\mathrm{kr}(A)$ is therefore not a change in the mathematical bound. [2]

The general inequality is formalized for the same index $i(A)$:

$$i(A) \le (D^2 - \mathrm{kr}(A) + 1)D^2.$$

The proof passes through the usual eventual full-rank formulation and a blocking argument, but the conclusion is the equality $S_n(A) = M_D(\mathbb{C})$ for one fixed length n. It is not merely a statement about the increasing sum $S_0(A) + \cdots + S_n(A)$.[3]

The two improved estimates are where the statements diverge. The formal invertible case assumes that one of the Kraus matrices, say A_{i_0}, is invertible; under that assumption it proves

$$S_{D^2-\mathrm{kr}(A)+1}(A) = M_D(\mathbb{C}).$$

The formal non-invertible case assumes that one Kraus matrix A_{i_0} is non-invertible and has a nonzero eigenvalue; under that assumption it proves

$$S_{D^2}(A) = M_D(\mathbb{C}).$$

The paper's hypotheses are weaker: they allow the invertible or non-invertible matrix to be any element of $S_1(A)$, hence any linear combination of the Kraus matrices.[4]

3 The mathematical difference

The distinction is small but genuine. Suppose first that the paper gives an invertible matrix $X \in S_1(A)$. Since X is a linear combination of the A_i, multiplication by X sends $S_n(A)$ into $S_{n+1}(A)$. Since X is invertible, this multiplication

[2] The formal definitions are MPSTensor.wordSpan in `TNLean/Wielandt/SpanGrowth/CumulativeSpan.lean:57--64` and MPSTensor.krausRank in `TNLean/Wielandt/Primitivity/Definitions.lean:84--90`. The local source identifies d as the number of linearly independent Kraus operators in `Papers/0909.5347/main.tex:360--365`; the proof of Lemma 1 uses $\dim T_1(A) = d$ in `Papers/0909.5347/main.tex:579--583`.

[3] The general theorem is MPSTensor.iIndex˙le˙general˙of˙isPrimitivePaper in `TNLean/Wielandt/Inequality/Bounds.lean:389--468`. The formal equivalence between the primitive condition used by Sanz–Pérez-García–Wolf–Cirac and eventual full Kraus rank under normalization is in `TNLean/Wielandt/Primitivity/Equivalence.lean:169--191`.

[4] The formal invertible case is MPSTensor.wordSpan˙eq˙top˙of˙isPrimitivePaper˙of˙isUnit and MPSTensor.iIndex˙le˙of˙isPrimitivePaper˙of˙isUnit in `TNLean/Wielandt/Inequality/Bounds.lean:313--337`. The formal non-invertible case is MPSTensor.wordSpan˙eq˙top˙of˙isPrimitivePaper˙of˙noninvertible˙eigenvector and MPSTensor.iIndex˙le˙sq˙of˙noninvertible˙eigenvector in `TNLean/Wielandt/Inequality/Bounds.lean:213--243`.

is injective on $M_D(\mathbb{C})$. This is the linear-algebra reason why the dimensions of the spaces $S_n(A)$ cannot stagnate for too long; it is the dimension-growth step used in the proof of the sharper bound.

If X is itself one of the Kraus matrices, then every product XW with $W \in S_n(A)$ is visibly a linear combination of words of length $n+1$ whose first letter is fixed. This is the case now covered by the formal theorem. If X is only a linear combination of Kraus matrices, the same inclusion $XS_n(A) \subseteq S_{n+1}(A)$ is still true, but it has to be used in that more linear form. The proof should not need the matrix X to be a distinguished letter; it only needs X to lie in $S_1(A)$ and to have the relevant spectral property.

The same issue appears in the non-invertible case. The source hypothesis gives a matrix $X \in S_1(A)$ which is singular but has a nonzero eigenvalue. The formal theorem proves the corresponding statement when this matrix is one of the Kraus matrices. The natural generalization is to run the same argument with an arbitrary such $X \in S_1(A)$, using the inclusion $XS_n(A) \subseteq S_{n+1}(A)$ rather than a distinguished first letter. Equivalently, one could prove a change of Kraus representation which places X among the displayed Kraus operators without changing the channel or the spaces $S_n(A)$ in the relevant way. Either approach would match the paper's statement.

4 What is not a discrepancy

The replacement of d by $\mathrm{kr}(A) = \dim S_1(A)$ is not a paper error and not a formal weakening. It records explicitly the linearly independent Kraus count used by the source proof.

Nor is the use of the increasing spaces

$$T_n(A) = S_0(A) + S_1(A) + \cdots + S_n(A)$$

by itself a theorem-level deviation. Some intermediate formal results prove $T_{D^2}(A) = M_D(\mathbb{C})$ under normality. These results are weaker than the full quantum Wielandt theorem, but the theorem in `TNLean/Wielandt/Inequality/Bounds.lean` proves the stated bound for $i(A)$, where $i(A)$ is defined using the spaces $S_n(A)$ themselves. [5]

Finally, the lower-level use of normality is not itself a mismatch. In these formal statements, normality is eventual full Kraus rank. Under the paper's normalization, the formal statements prove the equivalence between the paper's primitive condition and eventual full Kraus rank, so using normality as an intermediate hypothesis is legitimate.

[5] The cumulative-span theorem is MPSTensor.cumulativeSpan˙eq˙top in `TNLean/Wielandt/SpanGrowth/NonzeroTraceProduct.lean:164--197`; the theorem for the spaces $S_n(A)$ is the general theorem cited above. The blueprint makes this distinction in `blueprint/src/chapter/ch07_wielandt.tex:461--490` and states the general bound in `blueprint/src/chapter/ch07_wielandt.tex:492--548`.

5 Conclusion

The general quantum Wielandt bound has been formalized with the right index and with the paper's linearly independent Kraus count made explicit. The remaining difference concerns only the two improved estimates.

For those estimates, the paper assumes that the one-step space $S_1(A)$ contains a matrix with a specified spectral property. The formal statements presently assume that such a matrix is one of the chosen Kraus operators. This is a stronger hypothesis. Mathematically, the expected correction is clear: prove the two arguments for an arbitrary qualifying matrix $X \in S_1(A)$, using the inclusion $XS_n(A) \subseteq S_{n+1}(A)$ and the same dimension-growth reasoning, or prove an explicit Kraus-representation invariance result which reduces the paper's hypotheses to the chosen-Kraus-matrix hypotheses without changing the channel or the index $i(A)$.

References

[SPGWC10] M. Sanz, D. Pérez-García, M. M. Wolf, and J. I. Cirac. A quantum version of Wielandt's inequality. *IEEE Transactions on Information Theory*, 56(9):4668–4673, 2010.